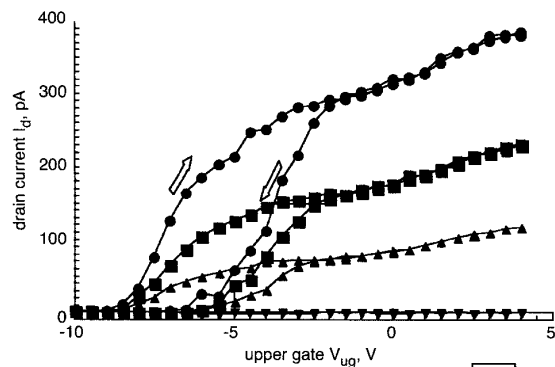


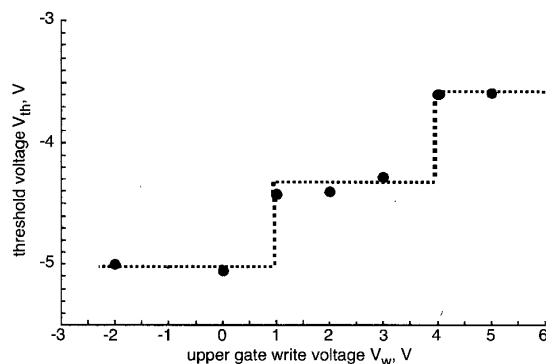
using the inorganic SiO<sub>2</sub> EB resist process. It should be noted that the ultra-narrow Si nanowire and side gates were simultaneously formed, since the SiO<sub>2</sub> layer acts as a positive-type EB resist with higher resolution than a conventional organic EB resist. Fig. 2 shows a plane-view TEM image of the Si nanodots made by RTO at 800°C for 20min of the 5.2nm thick a-Si:H film on the 1.9nm thick tunnel oxide. The TEM image confirms that crystalline Si nanodots with an average size of 3.6nm exist. The Si nanodot is small enough to observe the Coulomb blockade effect.



**Fig. 3** Drain current  $I_d$  against upper gate voltage  $V_{ug}$  characteristics at room temperature with drain voltage  $V_d$  as a parameter for Si nanodot nanowire memory transistor

Width, thickness and length of Si nanowire are 15nm, 20nm and 438nm, respectively. Round voltage sweep from -10V to 4V causes the threshold voltage shift  $\Delta V_{th}$  of 2.2V. Arrows indicate sweep direction

- $V_d = 200\text{mV}$
- $V_d = 150\text{mV}$
- ▲  $V_d = 100\text{mV}$
- ▼  $V_d = 0\text{V}$



**Fig. 4** Threshold voltage  $V_{th}$  against upper gate write voltage  $V_w$  at 13K for Si nanodot nanowire memory transistor

Note the remarkable single electron memory effect with large  $\Delta V_{th}$  of 0.72V and small  $\Delta V_w$  of 3.0V

- $V_d = 100\text{mV}$  measured
- $V_d = 100\text{mV}$  predicted

Electrical measurement of the fabricated Si nanodot nanowire memory transistor was performed at room temperature (RT) and at a low temperature. Fig. 3 shows the current hysteresis in the round-sweep drain current against upper gate voltage ( $I_d$ - $V_{ug}$ ) curves from -10V to 4V at RT. Note that a threshold voltage shift  $\Delta V_{th}$  of 2.2V occurs owing to electron charging in an Si nanodot on the channel. To further investigate the memory characteristic, the relationship between the threshold voltage  $V_{th}$  in the reverse sweep and the upper gate write voltage  $V_w$  was measured at a low temperature of 13K, as shown in Fig. 4. The step-like  $V_{th}$  shift is explained by a single electron charging effect of the Si nanodot. The large threshold voltage step  $\Delta V_{th}$  of 0.72V and the small write voltage step  $\Delta V_w$  of 3.0V indicate high single electron memory performance of the fabricated Si nanodevice. From the measured value of  $\Delta V_w$ , the Si nanodot radius is estimated to be 3.5nm. The Si nanodot size gives a single electron charging energy of 51meV. This value is fairly high compared with the thermal energy at RT.

It is concluded, therefore, that the  $\Delta V_{th}$  of 2.2V observed at RT shown in Fig. 3 is caused by charging of three electrons at the Si nanodot.

**Conclusion:** We have successfully fabricated an Si nanodot nanowire memory transistor with side gates using an inorganic SiO<sub>2</sub> EB resist process for the formation of the 15nm wide Si nanowire and an RTO process on an ultra-thin a-Si:H film to form an ultra-small Si nanodot. In the fabricated Si nanodevice, the remarkable single electron charging effect of  $\Delta V_{th} = 0.72\text{V}$  and  $\Delta V_w = 3.0\text{V}$  is confirmed at low temperature. The three electrons charging at the Si nanodot attribute a  $\Delta V_{th}$  of 2.2V observed at RT. The developed technology and measured electrical properties indicate the prospective future of Si nanodevices.

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## Variation in intrinsic parameters of transistor with gate width

M. Ariaudo, E. Bourdel and D. Pasquet

A method is developed that allows the intrinsic parameters of a transistor with a wide gate to be evaluated, taking the effects of propagation into account. It is shown that these parameters are not linearly related to the gate width. The proposed method provides a new transistor model that can be implemented in a circuit simulation.

**Introduction:** Much work is currently being devoted to the global analysis of circuits and their components [1, 2] because of the new phenomena that appear with the increase in frequency. Electrical simulation is no longer sufficient. For example, it is necessary to take into account the propagation effects on the conductors that modify the characteristics of the circuit or the intrinsic parameters of the components.

Some authors [3–6] have considered the propagation along the transistor width as if the dielectric beneath the conductors were a simple dielectric. Our method replaces this passive dielectric by a distributed transistor structure, using telegraphists' equations.

As a first step, electromagnetic simulation of the passive part of a field-effect transistor enables the conductors defined by the source, gate and drain fingers to be characterised. These parameters are introduced into a program that solves the equations of the coupled lines, taking the active part of the transistor into account. Other authors have considered two separate gate and drain modes [7–10]. We consider the two quasi-TEM modes together.

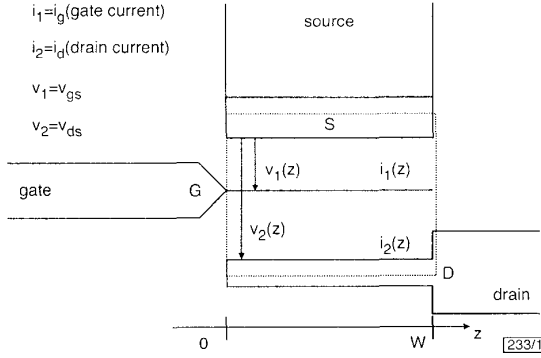


Fig. 1 Definition of voltages and currents

In Fig. 1, the metallic parts situated outside the dotted rectangle are simulated with a classical 2.5D electromagnetic simulation package (e.g. Momentum). The part inside the dotted rectangle is considered a coupled line structure. The connection points are shown as G, S and D in Fig. 1. They are considered as internal ports for the 2.5D electromagnetic simulator.

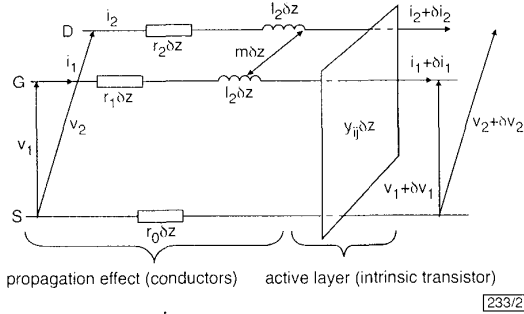


Fig. 2 Coupling of active and passive parts

Fig. 2 shows an equivalent scheme of a  $\delta z$  long coupled line. This represents a slice of a common source transistor. The series elements are computed by a 3D electromagnetic simulator (e.g. HFSS). The shunt elements are described by a unitary transistor.

If the transistor has several gates, only one pattern is simulated (Fig. 1).

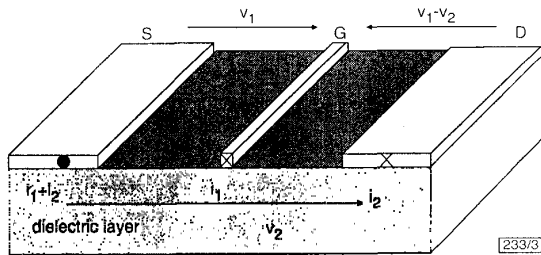


Fig. 3 Structure simulated with electromagnetic software

**Electromagnetic simulation:** The electromagnetic simulation takes into account the actual structure (Fig. 3), and the real characteristics of the electrodes. The symmetry of the structure is used by considering the structure in a common gate configuration. Two

modes can propagate: the coplanar and the slot modes. The results of the simulation are the impedances ( $Z_{copl}$  and  $Z_{slot}$ ) and the propagation constants ( $\gamma_{copl}$  and  $\gamma_{slot}$ ) of the modes. The elements characterising the structure must then be calculated.

According to the voltage and current conventions, the slot mode is characterised by the following conditions:

$$v_1 - v_2 = v_1 - (i_1 + i_2) = i_2 \quad \text{and} \quad \frac{\partial v_1}{\partial z} = -\gamma_{slot} Z_{slot} i_1$$

the coplanar mode by:

$$v_1 - v_2 = -v_1 \quad (i_1 + i_2) = i_2 \quad \text{and} \quad \frac{\partial v_2}{\partial z} = -\gamma_{copl} Z_{copl} i_2$$

From the voltage propagation equation, the series elements, characteristics of the propagation along the fingers, are

$$\begin{aligned} r_0 = r_2 &= 0.5 \operatorname{Real}(Z_{slot} \gamma_{slot}) \\ 4r_1 &= 4 \operatorname{Real}(Z_{slot} \gamma_{slot}) - \operatorname{Real}(Z_{copl} \gamma_{copl}) \\ 2m\omega &= \operatorname{Imag}(Z_{slot} \gamma_{slot}) \\ 4l_1\omega &= \operatorname{Imag}(Z_{slot} \gamma_{slot}) + 4 \operatorname{Imag}(Z_{copl} \gamma_{copl}) \\ l_2\omega &= \operatorname{Imag}(Z_{slot} \gamma_{slot}) \end{aligned}$$

Table 1: Parameters of unitary finger

| $C_{gs}$<br>(fF/ $\mu\text{m}$ ) | $R_i$<br>( $\Omega\mu\text{m}$ ) | $C_{gd}$<br>(fF/ $\mu\text{m}$ ) | $R_{gd}$<br>( $\Omega\mu\text{m}$ ) | $C_{ds}$<br>(fF/ $\mu\text{m}$ ) | $G_m$<br>(S/ $\mu\text{m}$ ) | $G_d$<br>(S/ $\mu\text{m}$ ) | $\tau$<br>(ps) |
|----------------------------------|----------------------------------|----------------------------------|-------------------------------------|----------------------------------|------------------------------|------------------------------|----------------|
| 0.9157                           | 555                              | 0.1166                           | 15180                               | 0.1263                           | $5.42 \times 10^{-4}$        | $3.82 \times 10^{-5}$        | 0.3            |

**Evaluation of equivalent parameters:** To couple the active and passive parts, we apply the coupled lines theory (Fig. 2). The series elements are deduced from the electromagnetic simulation. The  $y_{ij}$  shunt parameters are determined for the unitary intrinsic transistor.

This leads to an equations system the solution to which allows us to evaluate two propagation modes characterised by their propagation constants  $\gamma_1$  and  $\gamma_2$ ; the voltages and currents are the combination of these two modes. Only four boundary conditions are necessary to evaluate the solution:

$$\begin{aligned} i_1(W) &= i_2(0) = 0 \quad (\text{open lines}) \\ v_1(0) &= v_{gs} \quad v_2(W) = v_{ds} \end{aligned}$$

A one-finger transistor can be considered as a two-port characterised by its  $Y$  matrix. Its input and output voltages and currents are

$$i_1(0) = i_g \quad -i_2(W) = i_d \quad v_1(0) = v_{gs} \quad v_2(W) = v_{ds}$$

The  $Y$  parameters of this two-port are directly computed from the solution of the system

$$\begin{aligned} Y_{11} &= \left. \frac{i_1(0)}{v_1(0)} \right|_{v_2(W)=0} & Y_{12} &= \left. \frac{i_1(0)}{v_2(W)} \right|_{v_1(0)=0} \\ Y_{21} &= \left. \frac{-i_2(W)}{v_1(0)} \right|_{v_2(W)=0} & Y_{22} &= \left. \frac{-i_2(W)}{v_2(W)} \right|_{v_1(0)=0} \end{aligned}$$

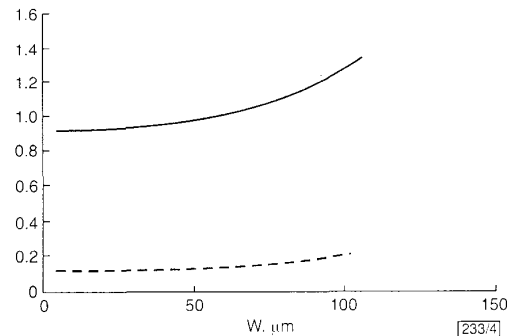


Fig. 4 Variation of capacities with gate width

**Application to PHEMT (EDO2H PML process):** We applied our method to a transistor characterised by the parameters in Table 1, for  $V_{gs} = -0.2V$  and  $V_{ds} = 3V$ .

The electromagnetic simulation was carried out at 40GHz.

As an example of the results, Fig. 4 shows the variation in the capacities deduced from the  $Y$  matrix against the gate width for a single gate transistor; the nonlinearity is clearly evident, which is the result of propagation along the fingers.

**Conclusion:** A method has been presented that enables the model of a wide transistor to be easily evaluated, taking into account the propagation effects along the fingers. This model can be used to design any circuit. At higher frequencies or for wider gates, the simulation shows new phenomena, which means that the experimental results would have to be compared in order to draw conclusions regarding the validity of the model and the interest of new configurations.

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## RNS quasi-chaotic generators

M. Panella and G. Martinelli

RNS digital filters are particularly suited to the realisation of quasi-chaotic generators. A very large number of architectures can be devised by relying on different moduli and poles. The proposed scheme is constituted by a cascade of first-order filters having poles coincident with first-order primitives.

**Introduction:** Digital filters with finite precision possess periodic autonomous responses, thereby precluding chaos. Only quasi-chaotic (QC) behaviour is thus possible, as suggested and defined in [1] on the basis of the results given in [2]. QC generators are particularly useful for realising encoders/decoders for secure communication, as shown in [1] and recently in [3]. A particular implementation of digital filters with finite precision may be based on the residue number system (RNS), resulting in filters that are automatically suited to the realisation of QC generators. Numer-

ous alternatives are possible in this regard by combining the periodic responses, which can be varied by relying on different moduli and poles. The use of poles coincident with primitive roots of different types (first, second and third order primitives [4]) yields responses which closely imitate, also if periodic, chaotic behaviour due to their spreading action. Moreover, it is very easy to obtain very large periods with the consequence of further pursuing the chaotic behaviour. In this Letter, we propose a QC generator based on first-order primitives.

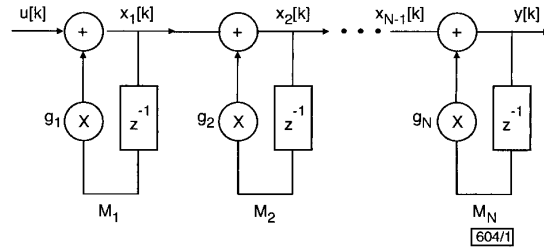


Fig. 1 Schematic diagram of proposed RNS QC generator

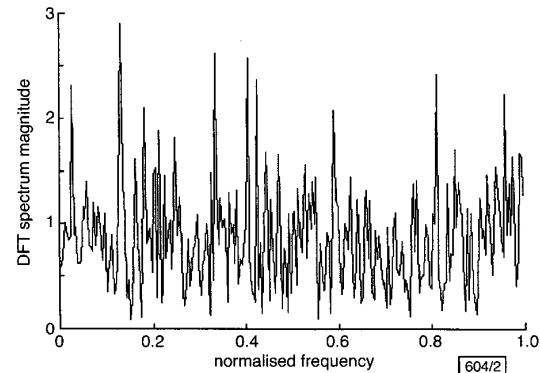


Fig. 2 DFT spectrum of one period of zero-input response

**RNS QC generator:** The RNS digital filter proposed as a QC generator is depicted in Fig. 1. It is composed of a cascade of  $N$  first-order RNS sections, each based on a prime modulus. The parameters of the filter are the number  $N$  of sections, the values of their moduli  $M_i$ ,  $i = 1, \dots, N$ , and the corresponding primitives  $g_i$  used as poles. With reference to the  $k$ th sample,  $k \geq 1$ , the output response  $y[k]$  can be computed using the following equations:

$$\begin{aligned} x_1[k] &= (g_1 * x_1[k-1] + u[k]) \bmod M_1 \\ x_2[k] &= (g_2 * x_2[k-1] + x_1[k]) \bmod M_2 \\ &\vdots \\ y[k] &= x_N[k] = (g_N * x_N[k-1] + x_{N-1}[k]) \bmod M_N \quad (1) \end{aligned}$$

where  $u[k]$  is the input and  $x_i[k]$ ,  $i = 1, \dots, N$ , is the output of the  $i$ th section. The filter can be considered an RNS QC generator in the absence of an input. Given the initial conditions  $x_i[0]$ ,  $i = 1, \dots, N$ , of the  $N$  sections, the resulting zero-input response will be the output of the QC generator. The zero-input response will be periodic, with a period coincident with the least common multiple of  $(M_i - 1)$ ,  $i = 1, \dots, N$ . The length of the period can be very large. For instance, considering the case

$$N = 8 \quad M_i = [257 \ 263 \ 347 \ 359 \ 383 \ 467 \ 479 \ 503] \quad (2)$$

the period will be

$$P = (256 * 131 * 173 * 179 * 191 * 233 * 239 * 251) \simeq 2.7 \times 10^{18} \quad (3)$$

The shape of the QC generator response does not depend on the initial conditions. The different responses obtained by varying them are simply shifted versions of its zero-input response. The shape of this response, instead, depends on the specific primitives  $g_i$ ,  $i = 1, \dots, N$ , used in the  $N$  sections. Each choice defines a particular 'state' of the generator. In the case of a prime modulus  $M$ ,